Manan Wadhwa

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Education

Maharaja Agrasen Institute of Technology (GGSIPU) — Delhi, India
B.Tech in Computer Science & Technology — *Expected Jul 2027*
BlueDot Impact — Technical AI Safety and Biosecurity certifications

Research & Professional Experience

MARS Fellow — Mentorship for Alignment Research Students — Jul 2026 – Present
Cambridge AI Safety Hub (CAISH) — *Oxford, UK / Remote*

- Built **model organisms of misalignment**: fine-tuned open LLMs to carry hidden objectives — **secret loyalties, inductive backdoors, and emergent misalignment** — inserted at different training stages (pretraining vs. SFT vs. RL-style post-training) to study which regimes yield the most **persistent** behaviors.
- Led **blue-team** evaluation separating behavioral masking from mechanistic erasure: used probing, activation steering, and **SAE feature analysis** to surface internal signatures of persistence, and tested whether "removed" behaviors resurface under mild retraining, distillation, and prompt variation.

Technical AI Safety Project — BlueDot Impact — Aug 2026
"Talk Does Not Come Apart From State Easily" — calibrating AI-welfare instruments | | *Certificate* — *Remote*

- Manufactured **ground truth** for AI-welfare instruments: LoRA fine-tuned **Qwen3 0.6B–32B** in a grid world whose rewards are never verbalised, installing organisms that avoid a tile with *no words* for it, talk aversively with the policy intact, or both — then screened six pre-registered instruments plus placebos for which axis each tracks.
- Across **five recipe families** and ∼260 builds (**DPO**-style contrastive supervision, **GRPO** on the remark), contingent narration proved hard to install without the state it describes ($r = 0.69$ with avoidance), while verbal instruments moved 5–10 logits under context and ≈0 under installed weights; released **44 adapters** and a **108-quantity reproduction script**.

LAMP Archaeological Pathfinding (HumanAI @ GSoC '26) | *IRL, Viewshed, GIS, NumPy* | May 2026 – August 2026

- **Google Summer of Code 2026 fellow;** implemented **7 routing algorithms** (A* + Tobler, MaxENT-IRL, spectral anomaly detection) over DEM data for the El Bagawat necropolis, Egypt, plus a **true 3D voxel viewshed** pipeline.

Technical AI Safety Fellow — Jan 2026 – May 2026
AI Safety Initiative (AISI) at Georgia Tech — *Remote*

- Studied **scalable oversight** (weak-to-strong generalization, debate), **machine unlearning** (WMDP, RMU) and **mechanistic interpretability** (superposition, circuits) with **Gemma Scope** and **interPLM SAEs**; red-teamed frontier LLMs with generated jailbreak prompts, and extended circuit analysis to protein LMs for biosecurity screening (below).

Publications & Research

Toxin Circuits in ESM-2: Mechanistic Interpretability for Biosecurity Screening — *GenBio @ ICML 2026*

- Showed ESM-2 linear probes detect **93.9%** of ProteinMPNN-redesigned toxins that fully evade BLAST (0%), establishing protein language model probes as viable biosecurity tools.
- Applied **SAE circuit analysis** (interPLM, 10,240 features, 205× compression) to mechanistically explain probe generalisation and the 93.9-point BLAST–ESM-2 gap; built a four-tier adversarial attack taxonomy bridging protein and LLM safety.

Getting in SHAPe: Error Detection through SHAP Attribution Geometry

- A logistic-regression meta-classifier over **8 SHAP dispersion features** detects misclassifications with no ground-truth labels: AUROC **0.843** (IMDB) / 0.786 (Adult Income), and 0.796 on high-confidence errors ($\hat{p} > 0.9$) where softmax is uninformative.

Selected Projects

FrameShift: Video Anomaly Detection Benchmark | *DINOv2, PatchCore, SAM 2* | — 2025

- Benchmarked **12 anomaly detectors** across 3 pipelines; implemented DINOv2 + PatchCore KNN with SAM 2 segmentation for fine-grained localization; ranked **6th of 1800+** in the Haas F1 Challenge.

LLM Fine-Tuning for Logical Reasoning | *Unsloth, LoRA, Llama 3.3-70B, ROCm* | — 2025

- Fine-tuned Llama 3.3-70B with **LoRA (rank 64)** on AMD MI300X (192 GB HBM3) — response-only loss masking, bfloat16 SFT — and built an adversarial **Q-Agent / A-Agent** framework with Qwen3-4B for automated MCQ generation.

Technical Skills

Languages: Python, C/C++, SQL, LaTeX
ML / DL: PyTorch, PyTorch Geometric, Hugging Face Transformers, Unsloth, timm, scikit-learn, NumPy
AI Safety & Alignment: mechanistic interpretability (SAEs / interPLM, circuits), probing & activation steering, model organisms of misalignment, AI-welfare instrument calibration, unlearning (RMU), scalable oversight, red teaming, protein LM biosecurity
Tools: Git, Docker, FastAPI, Weights & Biases, GDAL/GIS, CUDA, ROCm, Linux